

Answer Key — Interparticle Spacing, Compressibility and Diffusion

Final answer with a one-line reason. Full methods are in the Notes' worked examples.

Objective

- Q1** (c) **gases** — gases have the largest spaces between particles.
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- Q2** (a) **a gas** — large spaces make gases easily compressible.
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- Q3** (b) **diffusion** — this spreading is diffusion.
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- Q4** (c) **gases** — gases diffuse fastest due to large spacing and fast motion.
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- Q5** (a) **Both true, R explains A** — faster-moving particles spread faster — exactly as R says.
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- Q6** **compressed** — large spacing makes gases compressible.
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- Q7** **moving** — the constant motion of particles causes diffusion.
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Short answer

- Q8** Solids (smallest) < liquids < gases (largest).
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- Q9** Gases have large interparticle spaces so their particles can be pushed closer; liquids have little space, so they are almost incompressible.
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- Q10** The spreading of particles of one substance into another on their own; it is fastest in gases.
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Long / application

- Q11** (i) Gas particles have very large spaces and move freely in all directions, so they spread out to occupy the entire container. (ii) The food's particles diffuse through the air; because gas particles move fast and have large spaces, the smell spreads quickly to another room.
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- Q12** Fill a needle-less syringe with air, block the nozzle with your thumb and push the plunger — it moves in a lot (air compresses). Repeat with water — the plunger hardly moves. Air's large interparticle spaces let particles be pushed closer; water's particles are already close with little space, so it barely compresses.
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